

Haitao Xu

📍 2121 Euclid Ave, Cleveland, OH 44115

🏠 <https://haitaotsui.github.io> ✉ haitaotsui@gmail.com

EDUCATION

Cleveland State University, Cleveland, OH, USA

Ph.D. in Computer Science

Aug. 2021 - Aug. 2025

GPA: 3.84/4.00

Kunming University of Science and Technology, Kunming, China

M.S. in Software Engineering

Sept. 2011 - June 2014

GPA: 85.7/100

Shandong Institute of Commerce and Technology, Jinan, China

A.S. in Computer Science and Applications

Sept. 2005 - July 2008

GPA: 82.3/100

RESEARCH INTERESTS

Algorithms and Data Structures, Combinatorial Optimization, Operations Research, Theoretical Computer Science

PUBLICATIONS

Peer Reviewed Journal Articles

1. **Haitao Xu** and Jingru Zhang. The k-Center Problem of Uncertain Points on Graphs. *Information Processing Letters*, (2026): 106621.
2. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on cactus graphs. *Journal of Combinatorial Optimization*, 49.4 (2025): 54.
3. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on a real line. *Journal of Combinatorial Optimization*, 45.2 (2023): 68.
4. Qi Sun, Liwen Jiang, **Haitao Xu**. A Double-Layer Combination Algorithm for Real-Time Information-Sharing Network Design Problem. *Complexity*, 2021(1), 4856593.
5. Qi Sun, Liwen Jiang, **Haitao Xu**. Expectation-Maximization Algorithm of Gaussian Mixture Model for Vehicle-Commodity Matching in Logistics Supply Chain. *Complexity*, 2021(1), 9305890.
6. Song, Yang, Yan-Qiu Liu, Qi Sun, Ming-Fei Chen, and **Hai-Tao Xu**. A Joint Optimization Model considering the Product User's Risk Preference for Supply System Disruption. *Mathematical Problems in Engineering*, no. 1 (2021): 5081753.

Peer Reviewed Conference Papers

1. Qi Sun, **Haitao Xu**. A Unified Formal Verification for the k -Center Problem. *Accepted by the 11th International Symposium on Dependable Software Engineering: Theories, Tools and Applications (SETTA 2025)*. <https://www.setta2025.uk/accepted-papers>
2. **Haitao Xu** and Jingru Zhang. The two-center problem of uncertain points on trees. *In Proc. of the 16th Annual International Conference on Combinatorial Optimization and Applications (COCOA)*, pages 485-697, 2023.
3. Qi Sun, **Haitao Xu**. Adaptive Control Method of UAV Intelligent Rudder Based on Hybrid Genetic Algorithm. *In Proc. of the 33rd Chinese Control and Decision Conference (CCDC)*, pp. 4753-4757. IEEE, 2021.

4. Lijie Wu, Qi Sun, **Haitao Xu**, Xiaochen Song, Yang Zhang. Design of hybrid simulated annealing algorithm for UAV scheduling based on coordinated task scheduling. *In Proc. of the 40th Chinese Control Conference (CCC)*, pages 1669-1674. IEEE, 2021.

PROFESSIONAL EXPERIENCE (RECENT 10 YEARS)

1. Volunteer Research Assistant

Cleveland State University, Cleveland, Ohio, USA

Oct. 2025 - present

- Assisting with the research of designing algorithms for graph and geometry problems in facility locations. Preparing research papers and technical documentation.

2. Research Assistant/Teaching Assistant

Cleveland State University, Cleveland, Ohio, USA

Aug. 2021 - Aug. 2025

- Designed algorithms for facility location problems on line, tree, cactus, and general graphs, including two-center (line, tree, cactus) and k-center (general graphs) problems under uncertain demand data.
- Served as a Teaching Assistant for CIS550 Introduction to Algorithms, CIS606 Analysis of Algorithms, and CIS620 Advanced Operating Systems.

3. Software Engineer

KEDACOM, Shanghai, China

Apr. 2016 - Feb. 2020

Developed a video conferencing product consisting of both ALL-IN-ONE Terminal and Software Terminal.

- Built cross-platform components for Windows, Android, iOS, and Linux, enabling shared functionality and reducing redundant development across platforms.
- Extended an XMPP-based instant messaging module and designed reusable APIs with Protobuf, improving real-time communication reliability, integration efficiency, and cross-module performance.

4. Software Engineer

Shanghai Huateng Software Systems Inc., Shanghai, China

Aug. 2014 - Apr. 2016

- Developed a high-reliability, low-latency, high-concurrency debit card transaction system using C++, Tuxedo, Oracle/Informix, Bash, and Pro*C, ensuring robust and scalable transaction processing.
- Analyzed requirements, designed interfaces, and defined inter-system data formats for new services, while building batch settlement scripts and collaborating with cross-functional teams to enhance system integration, efficiency, and accuracy.

TECHNICAL SKILLS

1. Over 5 years of software development experience with a solid foundation in software analysis, design, system programming, and maintenance.
2. Experienced in software design and development using C/C++, Java, and Python, with hands-on experience in multithreading, data structures, algorithms, socket programming, and design patterns.
3. Experience with Oracle and MySQL, including high-performance SQL and database optimization.
4. Experience in web development using Java, JavaScript, HTML, CSS, NGINX and Docker.

PATENTS

1. Asynchronous multi-process service processing system, method, apparatus, and storage medium; No. CN109032767A

SERVICES

Journal Reviewer & Conference Reviewer

Hindawi Complexity; The 40th Chinese Control Conference (CCC'2021);